

CENG381 Stochastic Processes

Random Walks on Graphs — Practice 1

Q1 (30 points). Consider the star graph $K_{1,4}$ with a centre vertex C and four leaf vertices $\{L_1, L_2, L_3, L_4\}$. Every edge connects C to exactly one leaf; leaves are not connected to each other. The simple random walk at each step moves to a uniformly chosen neighbour.

- Write the 5×5 transition matrix P for the random walk (state order: C, L_1, L_2, L_3, L_4).
 - Using the formula $\pi(v) = \deg(v) / (2|E|)$, find the stationary distribution π for all five vertices.
 - Is the chain irreducible? Find the period of state C . Does the random walk converge to π ? If not, what is the standard fix?
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Q2 (35 points). Consider the 4-cycle graph C_4 with vertices $\{0, 1, 2, 3\}$ and edges $\{0-1, 1-2, 2-3, 3-0\}$. The simple random walk at each step moves to one of the two neighbouring vertices with equal probability.

- Write the 4×4 transition matrix P for the random walk on C_4 .
 - Find the stationary distribution π using $\pi(v) = \deg(v)/(2|E|)$. What is the period of state 0? Explain why the walk does NOT converge to π .
 - Construct the lazy walk $Q = (1/2) \cdot I + (1/2) \cdot P$. Write Q explicitly. Verify that state 0 has period 1 in Q , and confirm that π is still the stationary distribution of Q .
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Q3 (35 points). Consider the complete graph K_4 on vertices $\{A, B, C, D\}$: every pair of distinct vertices is connected by an edge. The simple random walk moves to each of the three neighbours with equal probability $1/3$.

- Write the 4×4 transition matrix P . Find π using $\pi(v) = \deg(v)/(2|E|)$. Is the chain irreducible? What is the period of state A ?
- The n -step return probability for vertex A is given by:
$$p_{\{AA\}}^{(n)} = 1/4 + (3/4) \cdot (-1/3)^n$$

Verify this formula for $n = 1$ and $n = 2$ by direct computation from the transition matrix.
- Using the formula from (b), find the smallest n for which $|p_{\{AA\}}^{(n)} - \pi_A| < 0.01$. Does the walk converge to π ? What drives this convergence even though K_4 has no self-loops?